



SIGN-OFF REQUIRED

This assessment must be digitally signed off by an Authorised User.

General Details

Assessor	Samuel Prieto Ayllon
Assessment Date	18/05/2026
Assigned Reviewer	Samuel Prieto Ayllon
Next Review Date	18/05/2027

Assigned user to Sign Off	Oraib AlKheetanRojas
Signed Off By	This assessment has not yet been signed off
Signed Off On	--
Sign Off Comments	

Operation Assessed	Charging rechargeable batteries
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Associated with specific area

-  Basement 2
-  B2 029 (Robotics Lab)
-  CTP

Description of work area and/or activity assessed

This is a risk assessment related to charging rechargeable batteries.

Reason for Risk Assessment

To identify, assess, and control health, safety, fire, thermal runaway, and equipment-damage risks associated with charging rechargeable batteries, including the use of chargers, power supplies, battery packs, charging areas, and safe storage before and after charging.

Overall Current Risk

Moderate Risk



Average Number of Persons Affected

2	 Contractors
4	 Employees
4	 Students

Current Rating	Hazard Information	Task Status
Moderate Risk 	Fire Incorrect charging parameters, incompatible chargers, damaged batteries, or poor charging practices may cause rechargeable batteries to overheat, short circuit, swell, ignite, or enter thermal runaway. A battery fire may produce flame, heat, re-ignition risk, and irritating or toxic fumes, creating risk of injury, fire spread, property damage, and equipment damage.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- Fire emergency procedures are in place and employees are briefed on them. - Fire alarm system, fire exits, exit signage, sprinkler system, and automatic fire/smoke detection are maintained and checked by the facilities team. - Fire extinguishers are provided and staff are trained in their use. - Batteries are charged only by trained/authorised users. - Batteries are charged only at the designated charging station using compatible chargers and correct battery-specific settings. - Batteries, chargers, leads, and connectors are inspected before charging; damaged, swollen, hot, leaking, crashed, water-damaged, unlabeled, or abnormal batteries must not be charged. - Batteries must not be left charging unattended, and overnight charging is not permitted unless separately approved. - Users periodically check for heat, swelling, noise, charger errors, or abnormal voltage during charging. - Batteries are disconnected after charging and returned to the correct storage container; batteries must not be stored on the charging station. - BAT-SAFE boxes, fire-resistant cabinet/OEM storage, metal tongs, sand bucket, alarm access, and exit routes are kept accessible.	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Low Risk 	 Gas/Fumes Rechargeable batteries, particularly lithium-based batteries, may release smoke, vapour, irritating or toxic fumes, electrolyte mist, or heavy odour if they are overcharged, damaged, overheated, short-circuited, incorrectly charged, or enter thermal runaway. Exposure to these fumes may cause eye, nose, throat, or respiratory irritation and may indicate an increased risk of fire or battery failure.	
Measures Currently in place to prevent risk of injury		
- Batteries are inspected before charging for swelling, deformation, leakage, odour, heat, damage, corrosion, water exposure, crash/impact history, or unknown status. - Damaged, swollen, leaking, hot, odorous, punctured, crashed, water-damaged, unlabeled, or abnormal batteries must not be charged. - Batteries are charged only by trained/authorised users using compatible chargers and correct battery-specific settings. - Charging is carried out only at the designated charging station. - Batteries must not be left charging unattended, and users periodically check for odour, vapour, smoke, swelling, heat, noise, charger errors, or abnormal voltage. - Charging must be stopped immediately if smoke, vapour, odour, swelling, overheating, or other abnormal behaviour is observed, where safe to do so. - People are moved away from the charging area if fumes, smoke, venting, or thermal runaway occurs. - Users must not touch or move hot, smoking, venting, hissing, sparking, or burning batteries. - Fire alarm/evacuation procedures are followed if there is smoke, fire, heavy odour, spreading smoke, or uncertainty. - Battery residue, contaminated sand, charger leads, or affected surfaces must not be handled after an incident until cleared by EHS/Public Safety.		

Potential Rating	Additional Controls Required
Low Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Moderate Risk 	 Electricity Incorrect chargers, damaged leads, exposed terminals, loose connectors, or contact with conductive surfaces may cause electric shock, short circuit, arcing, overheating, charger failure, or battery damage.	
Measures Currently in place to prevent risk of injury		
- Only compatible chargers and battery-specific charging settings are used. - Chargers, leads, connectors, balance boards, and batteries are inspected before use. - Damaged chargers, cables, connectors, or batteries must not be used. - Connectors and leads are kept away from conductive surfaces, including the charging table (covered by a non-conductive film). - Charging is carried out only by trained/authorised users. - Charging is stopped immediately if abnormal heat, arcing, charger errors, smoke, odour, or damaged wiring is observed.		

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.